VISVESVARAYA TECHNOLOGICAL UNIVERSITY

B.E. DEGREE EXAMINATION - SEMESTER IV (2023-24)

CS401 - DATABASE MANAGEMENT SYSTEMS

Time: 3 hours **Max marks: 100**

Note: Answer any FIVE full questions, choosing one full question from each module.

MODULE - 1

1a. Explain the concept of normalisation with a suitable example. Discuss 1NF, 2NF and 3NF with relation schemas. [08]

1b. Write SQL queries for the following: (i) list students with a CGPA above 8.5; (ii) find the department with the highest average salary. [08]

1c. Differentiate between a clustered and a non-clustered index. [04]

2a. Explain the concept of normalisation with a suitable example. Discuss 1NF, 2NF and 3NF with relation schemas. [08]

2b. Write SQL queries for the following: (i) list students with a CGPA above 8.5; (ii) find the department with the highest average salary. [08]

2c. Differentiate between a clustered and a non-clustered index. [04]

MODULE - 2

3a. Explain the concept of normalisation with a suitable example. Discuss 1NF, 2NF and 3NF with relation schemas. [08]

3b. Write SQL queries for the following: (i) list students with a CGPA above 8.5; (ii) find the department with the highest average salary. [08]

3c. Differentiate between a clustered and a non-clustered index. [04]

4a. Explain the concept of normalisation with a suitable example. Discuss 1NF, 2NF and 3NF with relation schemas. [08]

4b. Write SQL queries for the following: (i) list students with a CGPA above 8.5; (ii) find the department with the highest average salary. [08]

4c. Differentiate between a clustered and a non-clustered index. [04]

MODULE - 3

5a. Explain the concept of normalisation with a suitable example. Discuss 1NF, 2NF and 3NF with relation schemas. [08]

5b. Write SQL queries for the following: (i) list students with a CGPA above 8.5; (ii) find the department with the highest average salary. [08]

5c. Differentiate between a clustered and a non-clustered index. [04]

6a. Explain the concept of normalisation with a suitable example. Discuss 1NF, 2NF and 3NF with relation schemas. [08]

6b. Write SQL queries for the following: (i) list students with a CGPA above 8.5; (ii) find the department with the highest average salary. [08]

6c. Differentiate between a clustered and a non-clustered index. [04]

MODULE - 4

7a. Explain the concept of normalisation with a suitable example. Discuss 1NF, 2NF and 3NF with relation schemas. [08]

7b. Write SQL queries for the following: (i) list students with a CGPA above 8.5; (ii) find the department with the highest average salary. [08]

7c. Differentiate between a clustered and a non-clustered index. [04]

8a. Explain the concept of normalisation with a suitable example. Discuss 1NF, 2NF and 3NF with relation schemas. [08]

8b. Write SQL queries for the following: (i) list students with a CGPA above 8.5; (ii) find the department with the highest average salary. [08]

8c. Differentiate between a clustered and a non-clustered index. [04]

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MODULE - 5

9a. Explain the concept of normalisation with a suitable example. Discuss 1NF, 2NF and 3NF with relation schemas. [08]